

Executive Summary

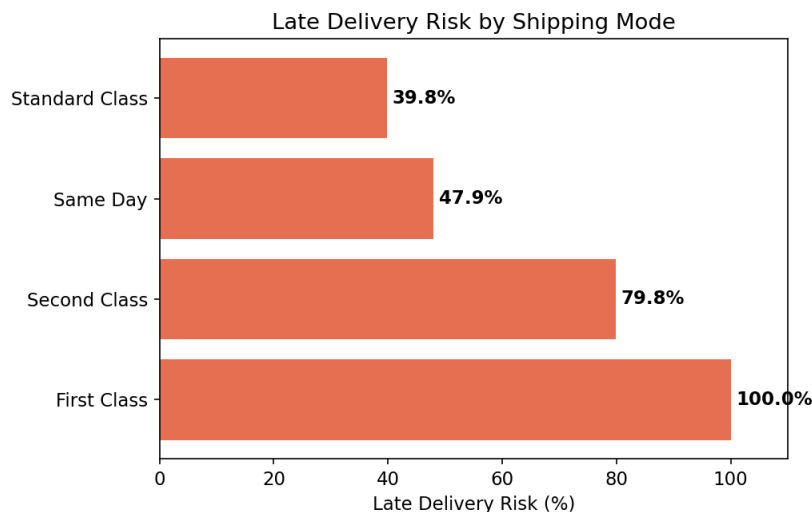
Delivery Performance & Delay Risk — APL Logistics (KWE Group)

This summary presents the key findings from a diagnostic analysis of 172,765 shipment records to assess delivery reliability across APL Logistics' global operations. The analysis was conducted to give stakeholders clear visibility into on-time performance and to identify the root causes of late deliveries before any predictive or corrective system is deployed.

Headline Result

Metric	Value
Orders Analyzed	172,765
On-Time Delivery Rate	18.6%
Late Delivery Rate	57.3%
Average Delay	+0.57 days

What Is Driving Delays



Delay risk is overwhelmingly explained by shipping-mode scheduling, not by geography or customer type. First Class shipments are late 100% of the time: they are promised in 1 day but consistently take 2 days to fulfill. This is a fixed, correctable scheduling mismatch rather than an unpredictable operational failure.

- First Class: 100% late (1-day promise vs. 2-day actual)
- Second Class: 79.8% late (2-day promise vs. 2–6 day actual spread)
- Same Day: 47.9% late
- Standard Class: 39.8% late — the best-performing mode, on a 4-day schedule

What Is Not Driving Delays

Region, market, and customer segment show only minor variation in delay risk (a narrow 55%–70% band across the board), in sharp contrast to the shipping-mode effect above. This means geographic or account-tier-specific interventions would have limited impact compared to fixing the shipping-mode scheduling mismatch.

Financial Impact

Delayed orders are approximately 4.8% less profitable per order than on-time orders (\$21.62 vs. \$22.71 average profit), reflecting the added cost of servicing SLA breaches and customer complaints.

Recommendations

- Re-align First Class SLA promises to 2 days, or invest in fulfillment capacity to genuinely support 1-day delivery.
- Review and extend the Second Class delivery window to reduce its 79.8% late rate.
- Prioritize shipping-mode fixes over regional programs — the data shows geography is a secondary factor.
- Add segment-aware monitoring so higher-value Corporate and Home Office accounts can be proactively flagged for delay risk.
- Use the accompanying Streamlit dashboard for ongoing monitoring, and treat this analysis as the baseline for future predictive delay-risk modeling.

Next Steps

This diagnostic analysis is the first stage of a broader program. The recommended next step is to pilot the First Class SLA correction, measure its impact on late_delivery_risk over one operating cycle, and expand successful changes to Second Class before investing in predictive modeling.